DARE: Differential Adversarial Reward Equalization for Humanoid Motion Imitation

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***Abstract*— Physics-based humanoid imitation must satisfy several heterogeneous objectives at once: global root motion, articulated pose, and velocity all have to agree with a synchronized reference. Handcrafted tracking rewards combine these objectives with weights and scales chosen per motion. Learned differential rewards remove the weights, but they keep the discriminator usable through regularization whose strength can affect both the critic's local geometry and the scale of its scores. We introduce DARE, a differential adversarial reward that fixes these two properties separately, with no motion-specific coefficient. The geometry of the critic is fixed by construction: every semantic block of the differential enters through its own spectrally normalized map, so objectives of very different dimension receive equal capacity and equal gain, and the score change relative to the unique zero-error anchor is bounded by the size of the residual. The scale of the critic is fixed once, in closed form, from the reward transform: the balanced adversarial loss places the anchor and the typical policy residual symmetrically around zero, and DARE rescales the critic so that this separation equals the width of the window over which the reward transform responds to the critic's score. The same closed-form calibration is used for every motion, with its scale computed automatically from the observed residual distribution. We evaluate DARE with one configuration across six dynamically diverse humanoid skills, achieving the lowest macro-average on all four geometric measures and the best motion-level behavior in the reported comparison, and deploy the resulting controllers on [TODO: robot platform].**

I. Introduction

Physics-based humanoid control must reproduce a prescribed reference motion under dynamics, actuation limits, and intermittent contact. Whether an imitation is accurate is not a single geometric question. The root has to travel and turn as the reference does, the articulated body has to take the same shape, joint and root velocities have to agree, and contacts have to occur at the right moment. These objectives are measured in different units, evolve on different time scales, and interact through the dynamics, so motion imitation is a multi-objective problem whose objectives cannot be pursued independently.

The standard response is to define a tracking term for each error and to combine the terms by hand. A representative scalar reward is

$$r_t^{\text{hand}} = \sum_{g=1}^{G} w_g \exp\Big[-\alpha_g \big\|\phi_g(\hat{s}_t) - \phi_g(s_t)\big\|_2^2\Big], \tag{1}$$

where ϕ_g extracts the g-th motion feature—joint rotations, root position, velocities, end-effector positions—from the simulated state s_t and the synchronized reference state $\hat{s}_t$. The weights w_g decide how objectives exchange against one another, and the scales α_g fix the numerical range of each error. When the motion changes, typical error magnitudes, learning difficulty, and dynamical roles change with it, so coefficients that serve one skill need not serve the next, and reward construction is repeated for every new motion.

Adversarial imitation offers a different route. Instead of declaring how many meters of root error are worth how many radians of joint error, a discriminator learns a scalar imitation signal from data. Motion priors such as AMP do this at the level of short state transitions and reward motion that resembles the dataset [7]. Adversarial differential discriminators make the tracking case precise: the discriminator receives the reference–policy differential and classifies whether it is the ideal zero vector, so the aggregation of tracking objectives is learned rather than specified [8]. The weights and scales of (1) disappear.

Existing differential adversarial formulations still rely on motion-dependent regularization to maintain a usable discriminator. The differential formulation has exactly one positive example, the origin, and without a suitable constraint a discriminator can become arbitrarily sharp: it scores the origin as ideal and gives nearly identical scores to the residuals visited by the policy. Regularization therefore affects the effective learning signal in two coupled ways. It changes the critic's local geometry and it changes the numerical range of the scores passed to the saturating reward transform. This coupling makes the effective regularization motion-dependent: as the residual distribution changes, the same setting can induce different critic geometries and score ranges.

DARE fixes each of these with no motion-specific coefficient. The geometry is fixed by construction, using two structural facts of the differential: perfect imitation is always the same point, $\Delta = 0$, whatever the motion or phase—we call this point the anchor—and the differential is a direct sum of semantic blocks—root position, root rotation, body position, body rotation, root velocities, joint velocities—which are the objectives of the tracking problem. Differential discriminators were conceived for general multi-objective optimization, where each objective is one scalar loss and the differential has one entry per objective, so all objectives enter on equal footing. The tracking instance replaces each entry by the raw coordinates of its objective, and the footing is lost: an 84-dimensional body-rotation block competes inside one dense layer with a 3-dimensional root-position block and wins by coordinate count. DARE restores the footing by mapping each block through its own spectrally normalized map to an equal-width latent—a learned, vector-valued generalization of the per-objective scalar loss—before a shared Lipschitz critic mixes the latents; the resulting critic changes its score, relative to the anchor, by no more than the size of the residual, for every motion and with no coefficient.

The scale is fixed by one calibration rather than by construction, because a Lipschitz bound says how steep a critic may be and not how steep it is. The balanced adversarial loss places the anchor score and the typical residual score symmetrically about zero, at $\pm M/2$ for a gap M that the loss drives as wide as the function class and the data allow. The saturating reward transform, however, responds to the score only inside a window of fixed width, and the two failure modes above are M overshooting or undershooting that window. DARE measures M once, when the input units become fixed, and rescales the critic so that anchor and typical residual sit at the two ends of the window. The rule is the same for every motion; only the measured separation, like the input normalization it follows, depends on the data.

The contributions are as follows. First, we identify that the regularization coefficient of differential adversarial rewards conflates critic geometry with critic scale, and we show that this is why it must be re-tuned per motion. Second, we introduce DARE, whose factorized, spectrally normalized critic fixes the geometry by construction and whose one-shot anchor calibration fixes the scale in closed form from the reward transform, so that no reward or regularization coefficient is specified per motion. Third, we evaluate DARE with one configuration across six heterogeneous skills and execute the resulting controllers on [TODO: robot platform].

II. Related Work

Physics-based motion tracking and learned imitation objectives. Physics-based character control has progressed from skill-specific, carefully engineered tracking objectives to learning-based systems that cover larger motion collections. DeepMimic combines pose, velocity, root, and end-effector tracking terms for physics-based skills [1]; subsequent work improves scalability and motion coverage through architectures, representations, and data [2]–[5]. Adversarial imitation provides a learned alternative to a manually written reward: GAIL matches expert and policy occupancies [6], while AMP learns a motion prior by discriminating short state transitions in physics-based characters [7]. These distribution-level objectives encourage plausible behavior, whereas reference-conditioned tracking requires agreement with a synchronized trajectory. ADD addresses this setting by presenting the reference–policy differential to an adversarial discriminator and learning a nonlinear aggregation of tracking objectives without component-wise reward weights [8].

Multi-objective learning and objective aggregation. Competing objectives are commonly combined by weighted scalarization, Pareto optimization, gradient balancing, adaptive loss weighting, or multiple objective critics [14]–[16]. These approaches differ in whether preferences are specified explicitly, adapted from objective statistics, or retained as separate signals. Adversarial differential learning offers another form of scalarization in which the aggregation is represented by a learned nonlinear function [8]. Motion tracking adds a representation issue to this problem: one physical objective can occupy a feature block with many more coordinates than another, even when both are conceptually single objectives.

Regularization of adversarial critics. Adversarial critics are regularized with gradient penalties, Lipschitz constraints, or spectral normalization to control sensitivity and improve optimization [9]–[11]. Gradient penalties constrain sampled input derivatives, whereas spectral normalization bounds the gain of individual network maps by construction. For non-Wasserstein adversarial losses, the numerical score range also determines the operating regime of the loss and its reward transform [12], [13]. Geometry and score calibration are therefore related but distinct considerations when a critic supplies the learning signal.

III. Background

A. Reference-Conditioned Humanoid Control

Let s_t be the simulated state at time t, $\hat{s}_t$ the synchronized reference state, and a_t the action, a vector of desired joint positions that a proportional–derivative servo converts into torques at the control frequency. A policy π_φ observes proprioception together with the next few reference frames and must close the loop under contact and underactuation; episodes begin at a random reference phase, so the policy has to recover the clip from any point. The policy maximizes the discounted return of a reward r_t with PPO [17], [18]; Section IV constructs r_t.

B. Motion Tracking as Multi-Objective Optimization

A motion descriptor decomposes a state into G feature groups $\phi_g : \mathcal{S} \to \mathbb{R}^{d_g}$, each describing one physical aspect of the motion: global root position, root rotation, root-relative body positions, body rotations, root linear and angular velocity, and joint velocity. A handcrafted reward first compresses each group into a distance $e_{g,t} = \|\phi_g(\hat{s}_t) - \phi_g(s_t)\|_2$ and then applies the fixed map (1). DeepMimic is a representative instance of this design: the relative weights and scales are specified before training and remain fixed while the policy visits its residual distribution [1]. A learned scalarization can instead adapt the trade-off to the residuals the current policy actually produces.

C. Learned Differential Objectives

Differential adversarial learning provides that scalarization [8]. In its general form, n objectives with non-negative losses $l^1, \ldots, l^n$ are collected into a differential vector $\Delta = (l^1, \ldots, l^n)$ that records the gap between ideal and actual performance on every objective. The ideal solution is the single point $\Delta = 0$. A discriminator $D(\Delta)$ trained to separate this point from the differentials of the current model is a learned, nonlinear aggregation of the objectives, and its gradient $\partial D / \partial l^i$ is the weight the aggregation currently assigns to objective i; that weight moves during training toward the objectives on which the model still falls short. For synchronized motion imitation the same construction is applied to the feature differential $\phi(\hat{s}_t) - \phi(s_t)$. Perfect tracking maps every pair onto one origin, so the positive class is again a single known point, and the differential retains information about *how* the current motion departs from the reference rather than collapsing it into a state occupancy. The policy is rewarded at each step with $-\log(1 - D(\Delta_t))$, as in AMP: large when the discriminator finds the residual Δ_t close to ideal, and vanishing when it does not.

A single positive point makes critic regularity important. Differential adversarial implementations therefore constrain the critic, for example with a sampled gradient penalty $\lambda\,\mathbb{E}_{\Delta\sim q_\pi}\|\nabla_\Delta D(\Delta)\|_2^2$ or with a Lipschitz-bounded parameterization, where q_π denotes the policy-residual distribution [8]. These controls shape how the critic responds over the sampled differential space; the numerical score range is a separate property of the learned function and of the reward transform.

IV. DARE: Differential Adversarial Reward Equalization

A. Learned Differential Reward

The distances $e_{g,t}$ are convenient for a weighted sum, but they discard the signed structure of the residual. Two policies can share the same root-position distance while drifting in opposite directions, and only one of them remains compatible with the next contact. DARE therefore keeps the differential itself. For each group g, a normalizer $\mathcal{N}_g = \mathrm{diag}(\nu_g)^{-1}$ rescales every coordinate by its mean absolute value ν_g, estimated from the residuals of the first 10^8 training samples and then fixed, as is standard for adversarial motion rewards. The normalizer has no offset, so it removes unit disparity between meters, rotation coordinates, and velocities while leaving the origin fixed; rotations use a tangent–normal embedding so that differences are Euclidean. The group-wise residual, the full differential, and the ideal point are

$$\Delta_{g,t} = \mathcal{N}_g\big(\phi_g(\hat{s}_t) - \phi_g(s_t)\big), \qquad \Delta_t = \bigoplus_{g=1}^{G} \Delta_{g,t}, \qquad \Delta^{+} = 0. \tag{2}$$

The vector $\Delta_t \in \mathbb{R}^d$, $d = \sum_g d_g$, retains error direction and cross-coordinate relationships instead of G distances, and every perfectly synchronized pair maps to the same origin regardless of the motion or its phase. Differentials produced by the current policy, together with recently visited differentials kept in a replay buffer, form the distribution q_π of non-ideal residuals.

DARE learns a scalar reward from this differential by training a critic to recognize the ideal point. Let $D_\theta(\Delta) = \sigma\big(f_\theta(\Delta)\big)$ be the probability that a differential is ideal, where σ is the logistic sigmoid and $f_\theta : \mathbb{R}^d \to \mathbb{R}$ is a scalar score. The origin is the ideal class and residuals drawn from q_π form the non-ideal class, so the critic is trained with

$$\max_\theta \ \log D_\theta(0) + \mathbb{E}_{\Delta\sim q_\pi}\big[\log\big(1 - D_\theta(\Delta)\big)\big]. \tag{3}$$

The first term accepts perfect imitation and the second rejects residuals from the current controller and its recent replay history. Maximizing (3) is equivalent to minimizing the balanced Bernoulli negative log-likelihood. Using $-\log\sigma(z) = \mathrm{softplus}(-z)$ and $-\log\big(1-\sigma(z)\big) = \mathrm{softplus}(z)$ gives

$$\mathcal{L}_{\mathrm{DARE}}(\theta) = \tfrac{1}{2}\,\mathrm{softplus}\big[-f_\theta(0)\big] + \tfrac{1}{2}\mathbb{E}_{\Delta\sim q_\pi}\,\mathrm{softplus}\big[f_\theta(\Delta)\big]. \tag{4}$$

The labels specify only whether a differential is ideal; they do not prescribe how root, pose, and velocity errors should be traded against one another. Their aggregation is represented by f_θ and adapts as the residual distribution q_π evolves during policy learning.

The critic score is converted into the policy reward as

$$r_t^{\mathrm{DARE}} = -\beta\log\big[1 - D_\theta(\Delta_t)\big] = \beta\,\mathrm{softplus}\big[f_\theta(\Delta_t)\big], \tag{5}$$

where the reward amplitude β is fixed and shared across all motions. Higher critic scores correspond to residuals judged closer to the ideal and therefore produce larger rewards. The same learned scalar function aggregates all tracking feature groups, without component-wise reward weights or success thresholds.

This formulation leaves two design requirements for the critic. First, feature groups with different dimensionalities should enter the learned aggregation on comparable footing rather than being favored by their representation size. Second, the numerical range of the critic score should remain compatible with the region in which (5) is informative. Sections IV-B and IV-C address these requirements separately through critic equalization and score calibration.

B. Equalized Critic: Geometry by Construction

We use "critic geometry" to mean how rapidly its score can change when a differential changes, namely its local sensitivity and global Lipschitz bound. In the general differential of Section III-C, each objective contributes one entry. Tracking instead exposes the d_g coordinates of each feature block Δ_g directly. After per-coordinate normalization, if coordinates have comparable typical magnitudes, the block norm $\|\Delta_g\|_2$ grows with $\sqrt{d_g}$. Consequently, a shared first layer can couple representation dimensionality to the sensitivity allocated to each semantic block. Let that layer be the dense map $W = [W_1 \ \cdots \ W_G]$ and let $\sigma_{\max}(W)$ denote its largest possible input amplification. Imposing $\sigma_{\max}(W) \le 1$ limits the gain of the complete map. Along any unit output direction u,

$$\sum_{g=1}^{G} \big\|W_g^\top u\big\|_2^2 = \big\|W^\top u\big\|_2^2 \le 1, \tag{6}$$

so the blocks share one overall sensitivity budget in each output direction. A dense layer therefore does not provide independent gain bounds for the semantic blocks; its effective allocation is entangled with the representation presented to it.

DARE preserves the vector differential while separating the semantic blocks before they interact. Each factor $\Delta_g \in \mathbb{R}^{d_g}$ is mapped by its own affine map and a 1-Lipschitz activation ρ to a latent representation of a common width, and the representations are concatenated:

$$\begin{aligned}\Phi_\oplus(\Delta) &= \bigoplus_{g=1}^{G} \psi_g(\Delta_g), \\ \psi_g(\Delta_g) &= \rho(\bar{W}_g \Delta_g + b_g), \qquad \bar{W}_g = \frac{W_g}{\sigma_{\max}(W_g)},\end{aligned} \tag{7}$$

where $W_g \in \mathbb{R}^{m \times d_g}$ and every factor receives the same latent width m. Each ψ_g is a scalar- or vector-valued representation of its block with a 1-Lipschitz bound; $m = 1$ gives a scalar-valued map, whereas $m > 1$ retains a vector representation and can preserve more of the block's residual structure. The direct sum keeps each representation in its own slot, so the next stage receives one equal-width entry per objective. A shared trunk then scores the concatenation, $g_\theta(\Delta) = \bar{T}(\Phi_\oplus(\Delta))$ with $\bar{T}(h) = \bar{w}^\top \rho(\bar{V}h + c) + b$, whose linear maps are spectrally normalized in the same way so that $\mathrm{Lip}(\bar{T}) \le 1$. We write g_θ for this raw critic output; the score f_θ that enters the loss (4) and the reward (5) is g_θ up to a single scalar fixed in Section IV-C, and equals g_θ until then. All affine maps carry a bias, so activation thresholds can sit where the residual distribution actually lives; the anchor $g_\theta(0)$ thus depends on every layer and is trained by the positive sample like any other point, while spectral normalization keeps each map non-expansive regardless of the offsets.

The direct-sum construction gives every objective the same latent width and an independent gain bound before interaction; it does not assert that the objectives have equal importance or equal residual norm. In contrast to the shared bound in (6), each factor in (7) has its own normalized map. Because the outputs are disjoint, their distances combine by a root-sum-square identity: for any two differentials,

$$\begin{aligned}\left\|\Phi_\oplus(\Delta) - \Phi_\oplus(\Delta')\right\|_2^2 &= \sum_{g=1}^{G} \left\|\psi_g(\Delta_g) - \psi_g(\Delta'_g)\right\|_2^2 \\ &\le \|\Delta - \Delta'\|_2^2,\end{aligned} \tag{8}$$

since each block has Lipschitz constant at most one. Thus a perturbation in one semantic subspace cannot be amplified through the coordinates of another before the shared trunk processes the slots. The trunk can still learn different sensitivities from the residual distribution, but this allocation is no longer forced by unequal input widths at the first map.

The unique ideal point now gives the geometric guarantee. Because every perfect pair occupies the origin, the quantity that governs reward variation is the score change relative to the anchor, and combining (8) with $\mathrm{Lip}(\bar{T}) \le 1$ gives

$$\left|g_\theta(0) - g_\theta(\Delta)\right| \le \|\Delta\|_2. \tag{9}$$

If the realized motion changes only slightly relative to the reference, the score changes by no more than the normalized residual. The biases cancel in score differences, so they move activation boundaries without loosening the bound. The parameterization therefore provides a motion-independent constraint on the score landscape. This is the geometric property needed before choosing the numerical score range; the next subsection addresses that scale separately.

C. Anchor Calibration: Scale in Closed Form

The bound (9) limits how steep the critic may be, but it does not determine the score range that the trained critic actually uses. The policy receives the score through the fixed transform (5), so this range determines how sensitive the reward is to tracking quality. We therefore measure the score gap between the ideal anchor and a typical policy residual. The reward's response to a score change is its slope,

$$\frac{\partial r^{\mathrm{DARE}}}{\partial f_\theta} = \beta\,\sigma(f_\theta), \tag{10}$$

a sigmoid that rises from 0 for very negative scores to β for very positive ones. Only over this transition does a score change produce a substantial change in reward. Far below it the reward is nearly zero even when the residual changes; far above it the transform is nearly linear and no longer provides a nonlinear shaping signal. The critic is most useful when the scores visited by the policy lie across this transition.

Where they fall is decided by the loss, not by the bound. The output bias of the critic shifts every score by the same amount, and the loss (4) is stationary with respect to that shift when its two terms have equal slope, $\mathbb{E}_{q_\pi}\sigma\big(f_\theta(\Delta)\big) = \sigma\big(-f_\theta(0)\big)$. The left-hand side is the mean probability of ideality the critic assigns to policy residuals, and the score whose sigmoid equals that mean is the natural representative of the negative class, $\bar{f}_\theta = \sigma^{-1}\big(\mathbb{E}_{q_\pi}\sigma(f_\theta(\Delta))\big)$, which we call the typical residual score. In these terms the stationarity condition reads $\bar{f}_\theta = -f_\theta(0)$: writing $M = f_\theta(0) - \bar{f}_\theta$ for the anchor gap,

$$f_\theta(0) = +\tfrac{M}{2}, \qquad \bar{f}_\theta = -\tfrac{M}{2}. \tag{11}$$

The anchor and the typical residual sit symmetrically about zero, and the loss pushes M as wide as the function class and the data permit. ($\bar{f}_\theta$ coincides with the arithmetic mean of the scores to second order in their spread; the runs reported below computed M with the arithmetic mean, which differed from $\bar{f}_\theta$ by TODO% at t_f.) The bound (9) caps M by the size of the residuals themselves; the realized value is some fraction of that cap, set by how much of the bound the trained network uses and by the residual scale of the motion, neither of which is a design quantity. The two failure modes of Section III-C are the two placements of M relative to the transition of (10). If M is too large, the typical residual at $-M/2$ lies far below the transition: its reward and its reward slope are both near zero, staying alive earns nothing, and improving a poor residual earns nothing either. If M is too small, the ratio of the reward slope at the anchor to that at the typical residual, $\sigma(M/2)/\sigma(-M/2) = e^{M/2}$, approaches one: the reward is close to the constant $\beta \log 2$ on every state, survival is rewarded, and the differences that distinguish good tracking from bad are compressed. Any regularizer that acts on sampled residuals can move M only indirectly through the critic's local slope. DARE instead sets the target score gap directly.

Both endpoints must therefore lie on the part of the transition where the slope is neither negligible nor saturated, and the question is how wide that part is. A sigmoid has no sharp edges; the conventional way to give its transition a width—the same convention used for the rise time of any sigmoidal response—is the span over which it climbs from 20% to 80% of its maximum. For the slope (10) this span is

$$w = \sigma^{-1}(0.8) - \sigma^{-1}(0.2) = \ln 4 - (-\ln 4) = \ln 16 \approx 2.77. \tag{12}$$

Placing the anchor at the upper end of the transition and the typical residual at the lower end means, by (11), $M = w$. The choice has a direct reading in terms of the critic itself: $D_\theta(0) = \sigma(\ln 4) = 0.8$ and $D_\theta(\bar{\Delta}) = 0.2$, so the calibrated critic assigns the anchor an 80% probability of being ideal and the typical residual 20%—a confident but not categorical classifier. A larger M drives it toward a hard classifier whose reward vanishes on typical residuals, a smaller M toward a coin flip whose reward carries no information; keeping an adversarial critic in a band of moderate confidence is standard practice [12]. The convention is fixed a priori and is not selected from task outcomes. Its empirical effect is assessed through the matched training curves and tracking metrics in Section V-C.

DARE achieves $M = w$ with one scalar applied to the raw critic g_θ of Section IV-B:

$$\begin{aligned} f_\theta(\Delta) &= \kappa\, g_\theta(\Delta), \qquad \kappa = \frac{w}{M_f}, \\ M_f &= g_\theta(0) - \sigma^{-1}\left(\mathbb{E}_{q_\pi}\sigma(g_\theta(\Delta))\right)\Big|_{t=t_f}. \end{aligned} \tag{13}$$

The gap M_f is measured once, at the iteration t_f at which the input normalizer $\mathcal{N}$ is fixed, over the same mixture of current and replayed residuals that forms the negative class of (4); κ is held constant thereafter, and before t_f it equals one. The moment is forced by the units: before t_f the coordinates of Δ are still being rescaled and M has no stable meaning. Both the loss (4) and the reward (5) use the calibrated f_θ, so the critic is trained and read out at one scale.

Three properties follow. The anchor is untouched: $\kappa \cdot 0 = 0$, and $f_\theta(0) = \kappa g_\theta(0)$ is still trained by the positive sample through every layer. The geometry is untouched: κ rescales the critic uniformly, so the ordering of residuals is preserved and (9) becomes $|f_\theta(0) - f_\theta(\Delta)| \leq \kappa\|\Delta\|_2$, a Lipschitz constant derived from the data rather than assumed to be one. And the rule contains no motion-specific number: w is a constant of the reward transform and M_f a statistic of the training data, so, like the normalizer $\mathcal{N}$, κ takes a different value on every motion because the same formula meets different residuals. Architecture, optimizer, replay, and reward amplitude remain fixed choices inherited from AMP and ADD; what DARE removes is the coefficient that had to be chosen per motion.

D. Training

Training alternates rollout collection, differential construction, critic updates, and PPO updates (Algorithm 1). Critic minibatches draw an equal mixture of current and replay differentials, with the ideal point presented once per minibatch as the positive class; at t_f, before the rewards of that iteration are computed, M_f is measured with the critic in evaluation mode and κ is set by (13). One critic, one reward, one policy, and one calibration rule make up the whole loop, and every quantity in it is shared across motions.

Algorithm 1 DARE Training

Require: reference motion $\mathcal{M}$, policy π_φ, value V_ω, raw critic g_θ, replay buffer $\mathcal{B}$, normalizer $\mathcal{N}$; $\kappa \leftarrow 1$

- **for** iteration $t = 1, 2, \ldots$ **do**
  - Collect a synchronized rollout with π_φ from random reference phases; construct Δ_t by (2); add to $\mathcal{B}$
  - **if** $\mathcal{N}$ was fixed this iteration **then**
    - $\bar{g} \leftarrow \sigma^{-1}\left(\mathbb{E}_{\mathcal{B}\cup\text{rollout}}\,\sigma(g_\theta(\Delta))\right)$
    - $M_f \leftarrow g_\theta(0) - \bar{g}$; $\quad \kappa \leftarrow \ln 16/M_f$
  - **end if**
  - Compute r_t^{DARE} by (5) with $f_\theta = \kappa g_\theta$
  - **for** critic minibatch **do**
    - Sample equal numbers of current and replay differentials; update g_θ by one step on (4) with $f_\theta = \kappa g_\theta$
  - **end for**
  - Update π_φ and V_ω with PPO on r_t^{DARE}
- **end for**

TABLE I

DARE CONFIGURATION SHARED ACROSS ALL SIX SKILLS.

Component	Setting
Differential	single frame, $d = 172$
Semantic factors	G=7: root/body pos. (3/45), root/body rot. (6/84), root lin./ang. vel. (3/3), DoF vel. (28)
Direct-sum layer	$m = 146$ per factor (1022 total)
Shared trunk	linear $1022 \to 512$; output $512 \to 1$
Activation ρ	ReLU (1-Lipschitz)
Spectral norm.	all linear maps, one power iteration
Normalizer $\mathcal{N}$	per-coordinate mean-abs scale, fixed after 10^8 samples (iter. 382)
Logit calibration	$\kappa = \ln 16/M_f$, once at iter. 382
Critic optimizer	SGD, lr 2.5×10^{-4}, momentum 0.9
Critic minibatch	16384 current + 16384 replay
Replay buffer	2×10^5 differentials
Reward amplitude β	2

Calibration rule shared; resulting κ is data-dependent.

V. Experiments

A. Experimental Setup

All methods are evaluated using the same 28-DoF, 15-body humanoid model adopted in prior physics-based motion-imitation studies. Simulation runs in Isaac Lab at 120 Hz, with policy actions applied at 30 Hz through PD joint-position control [19], [20]. All four methods are trained under the same simulation environment, training budget, motion clips, random-phase initialization, and evaluation protocol. Each policy is trained for 2000 iterations of 8192 environments $\times$ 32 steps (5.24×10^8 samples), with five independent seeds, and is evaluated on 4096 episodes per seed from randomly sampled reference phases. Actor and value networks are two-layer MLPs (1024, 512), and the PPO settings (clip 0.2, GAE 0.95, $\gamma = 0.99$, SGD with momentum 0.9) are fixed across methods. DARE additionally uses the normalizer and one-shot logit calibration specified in Table I;

TABLE II

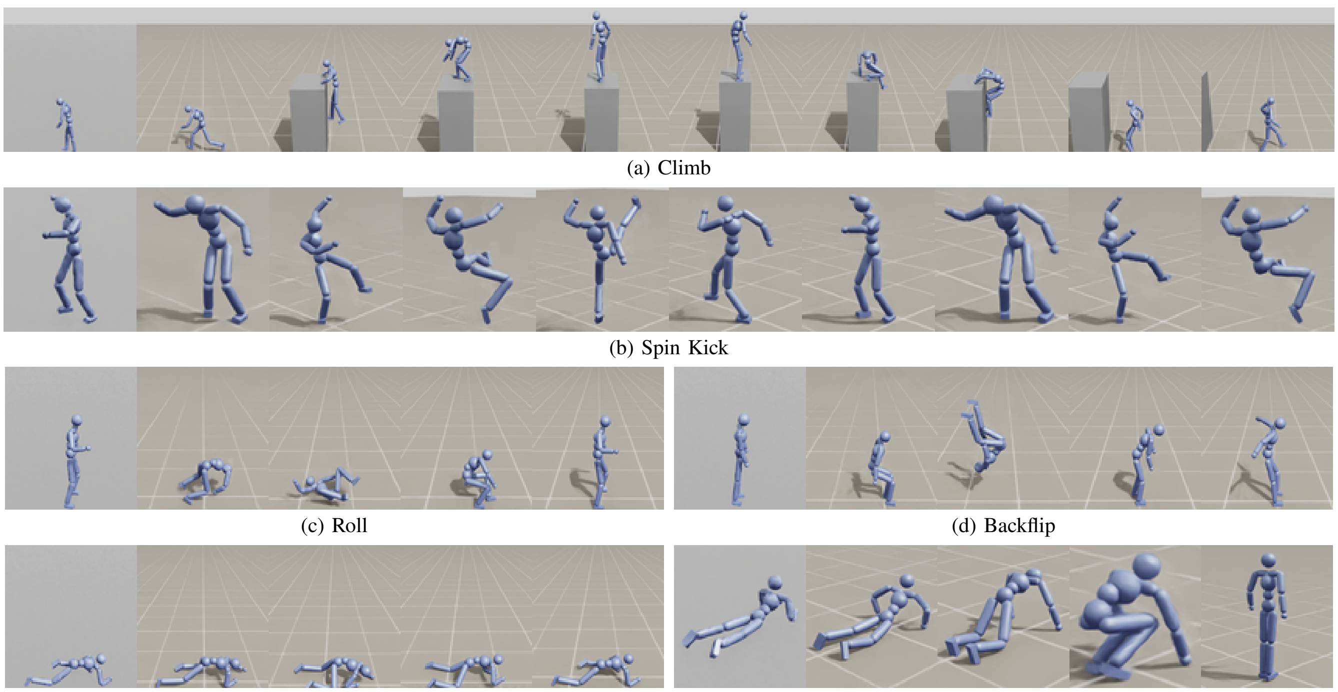
Macro-average tracking and motion-level behavior under the matched training and evaluation protocol. Entries are mean ± standard deviation over five seeds and six skills. Lower is better except for Foot Progress, which is ideal at one. DARE uses the same configuration on all skills.

(a) Geometric tracking

Metric ↓	AMP	DeepMimic	ADD	DARE
Root Pos. [m]	0.9116±0.084	0.0703±0.012	0.0725±0.010	**0.0528±0.008**
Body Pos. [m]	0.2169±0.031	0.0557±0.009	0.0363±0.007	**0.0313±0.006**
Root Rot. [rad]	0.7311±0.076	0.1459±0.021	0.0673±0.011	**0.0529±0.009**
Body Rot. [rad]	0.8098±0.069	0.2145±0.028	0.1358±0.017	**0.1246±0.013**

(b) Motion-level behavior

Metric	AMP	DeepMimic	ADD	DARE
Foot progress → 1	1.1335±0.046	1.0060±0.018	1.0109±0.021	**0.9989±0.012**
Lateral drift ↓	0.1127±0.019	0.0277±0.007	0.0435±0.009	**0.0171±0.005**



(a) Climb

(b) Spin Kick

(c) Roll

(d) Backflip

(e) Crawl

(f) Get-Up

Fig. 1. DARE policies executing the six skills in Isaac Lab. Panels use a consistent front-view crop; Climb and Spin Kick are shown with ten panels and the four other skills with five. All panels use the same fixed camera and aspect ratio. All controllers use the shared DARE configuration in Table I.

the normalizer is fixed after 10^8 samples (iteration 382).

The benchmark contains six skills with substantially different motion durations, contact patterns, and dynamical characteristics: Roll, Climb, Backflip, Crawl, Get-Up, and Spin Kick. Training termination follows each method's environment configuration (pose termination is enabled for DeepMimic, ADD, and DARE and disabled for AMP). The evaluation harness accumulates each metric over the executed reference-aligned steps and uses the same aggregation rule for every method; terminated trajectories are not padded with synthetic states. For the macro-averages in Table II, the six skills contribute equally before the five-seed statistics are computed.

We use six complementary measures. Four describe geometric tracking at two spatial levels: root position and root rotation measure global translation and orientation, while body position and body rotation measure articulated pose fidelity after removing the corresponding root motion. For an evaluation trajectory with B bodies, where r, p_b and q, q_b denote simulated root, body positions and orientations and hats denote reference values, these quantities are

$$\begin{aligned} E_{\text{root-pos}} &= \|\hat{r} - r\|_2, \\ E_{\text{root-rot}} &= \angle(q, \hat{q}), \\ E_{\text{body-pos}} &= \frac{1}{B}\sum_{b=1}^{B} \|(\hat{p}_b - \hat{r}) - (p_b - r)\|_2\,, \\ E_{\text{body-rot}} &= \frac{1}{B}\sum_{b=1}^{B} \angle(q_b, \hat{q}_b). \end{aligned} \tag{14}$$

where $\angle(\cdot,\cdot)$ is the shortest angle between normalized quater-

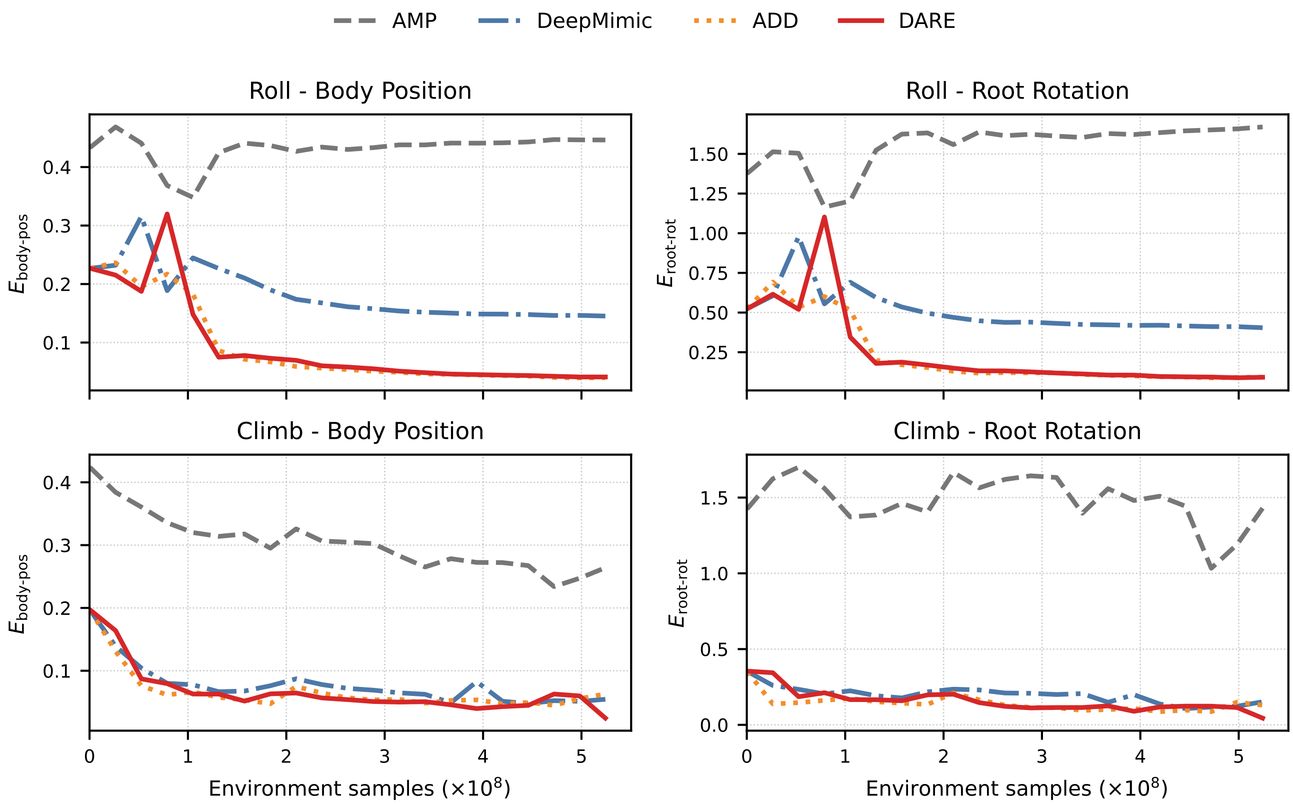


Fig. 2. Five-seed mean learning curves for Roll and Climb under the matched training protocol. The plotted curves show the mean over five independent seeds. The four panels pair body-position and root-rotation errors, exposing translational and orientation fidelity for short and long, contact-rich skills. The macro-average summary is reported in Table II.

nions. The remaining two measures describe motion-level behavior. Let Δr and $\Delta\hat{r}$ be the simulated and reference root displacements over the evaluated trajectory, and let $\Delta p_j, \Delta\hat{p}_j$ be the corresponding displacements of foot j. The reference-projected progress and transverse drift are

$$\begin{aligned} \rho_{\text{foot}} &= \min_j \frac{\Delta p_j^\top \Delta\hat{p}_j}{\|\Delta\hat{p}_j\|_2^2}, \\ \rho_r &= \frac{\Delta r^\top \Delta\hat{r}}{\|\Delta\hat{r}\|_2^2}, \\ E_{\text{lat}} &= \frac{\|\Delta r - \rho_r \Delta\hat{r}\|_2}{\|\Delta\hat{r}\|_2}. \end{aligned} \tag{15}$$

Foot progress is ideal at one, while lateral drift is minimized at zero. All six quantities are averaged over the evaluation trajectories; lower is better except for foot progress, whose target value is one.

B. Cross-Motion Tracking

Table II asks whether one configuration can serve skills whose contacts, horizons, and velocity profiles differ, and Fig. 1 shows the resulting controllers executing the six skills in Isaac Lab. Using one shared configuration, DARE is evaluated against the three baselines under the matched protocol of Section V-A. Table II reports the macro-average geometric errors and motion-level behavior across the six skills; DARE has the lowest value in every geometric metric and the closest-to-one foot progress ratio, while also giving the smallest lateral drift.

C. Learning Dynamics

Figure 2 compares the five-seed mean body-position and root-rotation errors against environment samples under a matched budget. DARE reaches a lower plateau than the baselines on both skills while retaining the same trend across the short Roll and contact-rich Climb clips. The curves therefore complement the final-checkpoint comparison in Table II.

D. Hardware Deployment

[TODO: Report the robot platform and its degrees of freedom, reference retargeting, domain randomization, observation noise, action filtering, hardware control frequency, and the number of trials and successful executions for each skill. State that the shared DARE reward formulation and calibration rule are used without motion-specific modification.]

VI. Conclusion

Manual tracking rewards turn the aggregation of heterogeneous imitation objectives into per-motion engineering. Learned differential rewards remove the component-wise weights, but regularization can still couple the critic's geometric sensitivity to the score range seen by the policy. DARE treats these as separate design quantities: the factorized critic specifies the geometry, and one closed-form calibration sets the score scale without a motion-specific coefficient. Every objective enters through its own spectrally normalized slot, so a three-dimensional root residual and an eighty-four-dimensional rotation residual stand on equal footing and the score change relative to the unique zero differential is bounded by the residual; the critic is then rescaled once, in closed form, so that the anchor and the typical residual sit at the two ends of the reward slope's transition. Across the six skills, DARE attains the lowest macro-average for all four geometric errors, the foot-progress ratio closest to one, and the smallest lateral drift in Table II. Extending DARE to multi-clip datasets and to other morphologies requires no change to the rule, and the factorized critic offers a direct place to attach task objectives as further semantic blocks.

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